WEB APPENDIX

In the Company of Strangers:

Social Influence from Anonymous Peers in Online Game Settings

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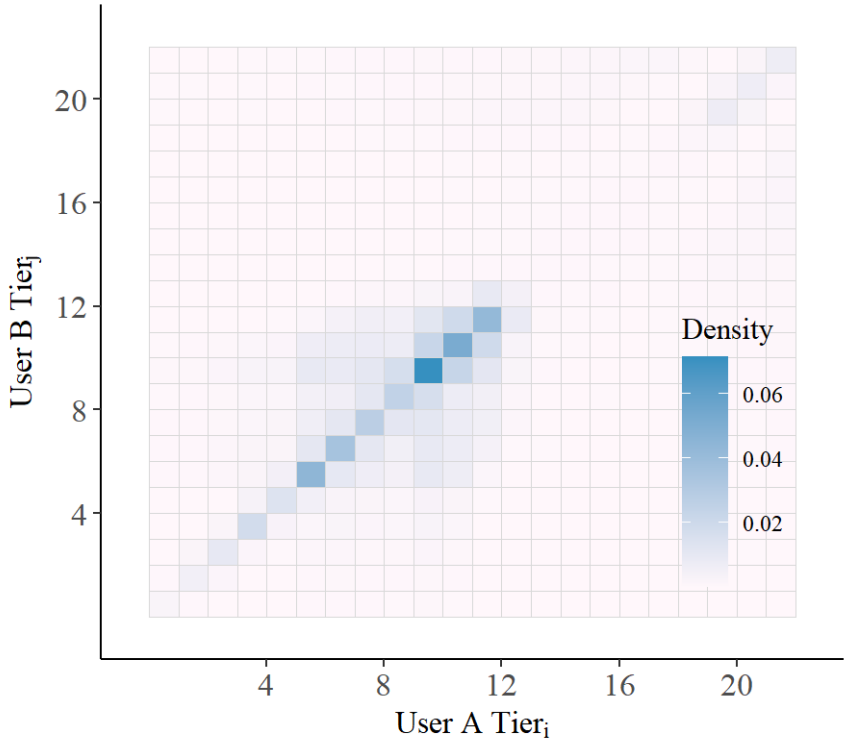
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WEB APPENDIX A: Additional Information on Social Match Setups

A1. Potential Imbalance in Match Formation

Figure A1: Social Match Set-ups by User Expertise—Heat Map



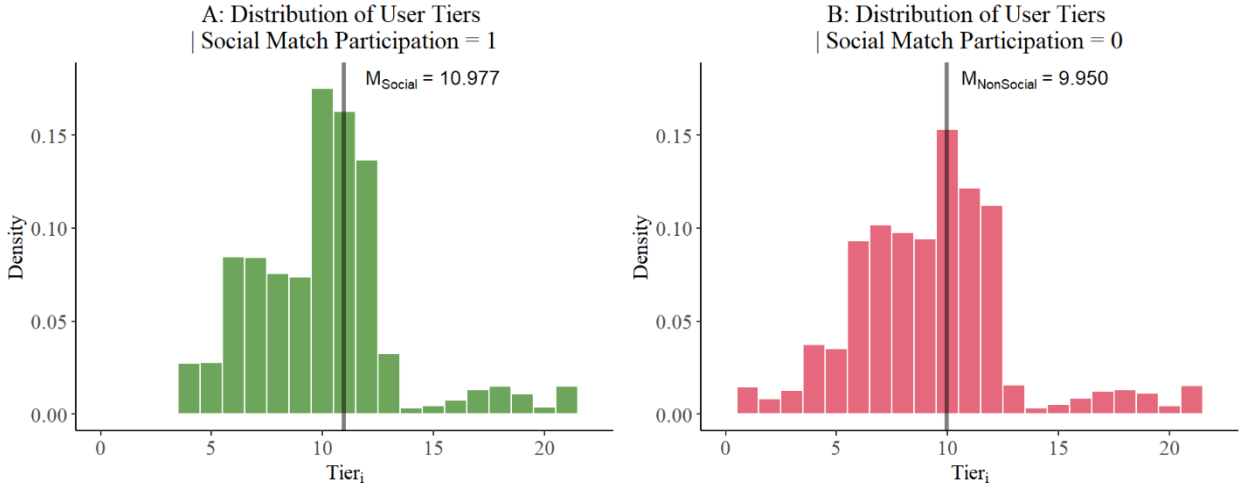
To illustrate the social match setups in Figure 2, we plot a heatmap-style representation with a density legend. This figure further suggests the potential endogenous formulation of social matches by user tier (45-degree line). We address this issue by using non-overlapping peers in an intransitive peer network (refer to our METHODOLOGY section for details).

A2. User Tier Distribution by Weekly Social Match Participation Status

Figure A2 illustrates the overall pattern of user tiers in a user-week panel dataset (original sample of 4,000 users), conditioned by social match participation. Notably, while the overall shape of the distribution appears similar, the subset of user tiers with no social match participation (Figure A2.B) shows a greater concentration of observations in the lower tiers (below five).

This pattern supports our cautious empirical approach, suggesting that there could be potential selection issues related to social match participation. As noted in the manuscript, we address this issue directly by explicitly modeling social match participation through an IV approach.

Figure A2: Breakdown of Tier Distribution by Social Match Participation



Note: Color is available online. Vertical lines denote the mean values of distributions.

WEB APPENDIX B:

Use of Control Function to Interaction Term with Endogenous Variable

The control function approach offers advantages over the standard instrumental variable approach, particularly when dealing with interactions (Papies, Ebbes and Van Heerde 2017; Wooldridge 2015). The inclusion of the control function offers a robust method to address potential endogeneity in both the primary regressor and its interaction terms. Theoretically, the origin of the potential endogeneity in the interaction term can be traced back to the main endogenous variable. This is because the interaction term is essentially a product of the main variable and another variable. If the main variable is endogenous, this endogeneity will naturally propagate to the interaction term.

However, the control function, by construction, captures and corrects for the endogenous components of the main variable. Consequently, when the control function adjusts for the endogeneity in the main variable, it also inherently corrects for the endogeneity present in the interaction term. Thus, the control function does not need to be interacted with moderators (See also Han, Mittal and Zhang 2017; Shen and Xiao 2014). A detailed mathematical illustration of this concept is provided below.

As in our context, suppose that we are interested in estimating β in the following equation:

$$y_{it} = \alpha_i + \tau_t + \beta x_{it} + \gamma \mathbf{Z_{it}} + \varepsilon_{it} \qquad \text{B.(1)}$$

- y_{it} is the spending observed for user i in week t.
- α_i is the user fixed effect.
- τ_t is the week fixed effect.
- x_{it} is the peer exposure for user i in week t-1. (potentially endogenous)
- $\mathbf{Z_{it}}$ is a vector of control variables.
- ε_{it} is the error term.

Now, suppose we identify a strong and valid instrument—in our case, average purchases of peers of $\text{peers}_{\text{it-2}}$—and estimate the first-stage regression. Specifically, we first identify all the non-overlapping peers of focal player i using the social match logs of each week. That is, we trace back to the peers of the focal player in week t-1. Then, we find the **other peers that these peers interacted** with in week t-2, find their purchases in week t-2, and take the average values. The specific results of this estimation are provided in Table C2 of Web Appendix C.

From this regression, we derive a control function, or residual, by subtracting the fitted value $(\hat{x}_{it})$ from the actual value(x_{it}).

$$\hat{V}_{it} = x_{it} - \hat{x}_{it} \qquad \text{B.(2)}$$

The completed second-stage model involves adding the control function $(\hat{V}_{it})$ as an additional regressor to the main model. This can be expressed as follows:

$$y_{it} = \alpha_i + \tau_t + \beta x_{it} + \gamma \mathbf{Z_{it}} + \theta \hat{V}_{it} + \varepsilon_{it} \quad \text{B.(3)}$$

$$y_{it} = \alpha_i + \tau_t + \beta x_{it} + \gamma \mathbf{Z_{it}} + \theta (x_{it} - \hat{x}_{it}) + \varepsilon_{it} \quad \text{B.(4)}$$

$$y_{it} = \alpha_i + \tau_t + (\beta + \theta) x_{it} + \gamma \mathbf{Z_{it}} - \theta \hat{x}_{it} + \varepsilon_{it} \quad \text{B.(5)}$$

The final equation (5) indicates that when the control function is included, the estimated coefficient of x_{it} changes from β to $(\beta + \theta)$. Empirically, θ is the estimated coefficient of the control function, demonstrating that θ directly adjusts the magnitude of the estimated effect of the endogenous regressor.

Now, suppose we introduce an additional regressor, h_{it}, and create an interaction with the endogenous regressor. For example, this new regressor could be the variables we use to test underlying mechanisms, such as contextual knowledge or threats to status. The equation then evolves as follows:

$$y_{it} = \alpha_i + \tau_t + \beta x_{it} + \psi h_{it} + \phi x_{it} h_{it} + \gamma \mathbf{Z_{it}} + \theta \hat{V}_{it} + \varepsilon_{it} \quad \text{B.(6)}$$

An important question here is if the inclusion of a control function accounts for potential endogeneity in the interaction term $x_{it} h_{it}$. The equation can be reformulated as follows:

$$y_{it} = \alpha_i + \tau_t + (\beta + \phi h_{it}) x_{it} + \psi h_{it} + \gamma \mathbf{Z_{it}} + \theta \hat{V}_{it} + \varepsilon_{it} \quad \text{B.(7)}$$

Here, the term $(\beta + \phi h_{it})$ captures the combined effect of the main influence of x_{it} and its interaction with h_{it}. If we further reformulate the equation, it becomes:

$$y_{it} = \alpha_i + \tau_t + (\beta + \phi h_{it} + \theta) x_{it} + \psi h_{it} + \gamma \mathbf{Z_{it}} - \theta \hat{x}_{it} + \varepsilon_{it} \quad \text{B.(8)}$$

Here, it is clear that the combined influence of both the main effect and the interaction, represented by $(\beta + \phi h_{it} + \theta)$, is adjusted by θ in the same manner as observed for the main effect.

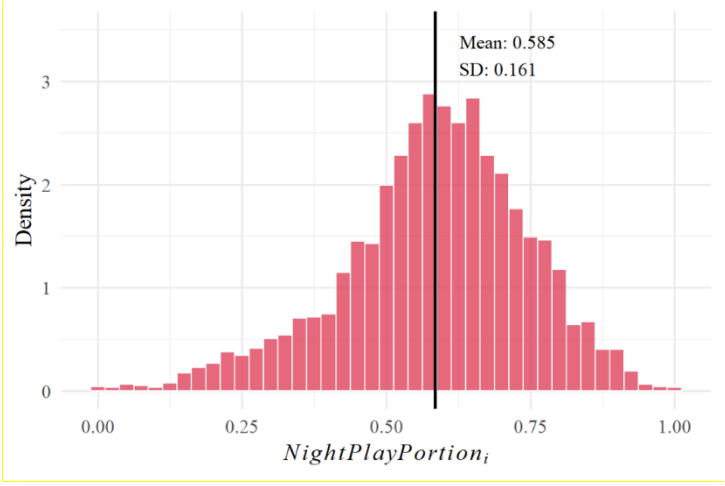
WEB APPENDIX C: Results of First-stage Regressions

C.1. Selection Model of Social Match Participation

We present the empirical results of the first-stage regression on social match participation using an instrumental variable (IV). Our IV consists of two components: $NightPlayPortion_i$, which computes the ratio of total gameplay minutes during the night (from 6:00 PM – 6:00 AM) to the total gameplay minutes, and $ServerDown_t$, a binary indicator of unexpected game server outages for maintenance.

Potential concerns in the use of this IV include: (1) the possibility of a degenerate condition where the IV has no variation at all during the entire observation period. This could occur if a user has a zero value in $NightPlayPortion_i$. However, we find this is unlikely because only 3 out of 3,699 players (.08%) have an exactly zero value for this variable. Figure C below depicts the distribution of this cross-sectional variation among the sample, with a mean of .585 and a standard deviation of .161, as shown.

Figure C: Distribution of $NightPlayPortion_i$ Variable



Note: Color is available online. The vertical bar denotes the position of the mean value.

Another concern may be that $NightPlayPortion_i$ is at least partially endogenous to the error term. Specifically, the propensity to play the game at night may be correlated with how much money each player spends weekly. However, this concern is substantially mitigated because, in the first-stage regression, we use a random effects Probit model, where player-level random effects absorb the influence of such unobserved factors.

The more explanation of the IV and the identification assumptions are discussed in the METHODOLOGY section. Here, we report the detailed results for (1) the relevance and (2) the exclusion restriction conditions of the IV for identifying the effect of interest.

Relevance. To assess the empirical strength of our instrumental variable (IV), we estimate a first-stage random effect Probit model using the social match participation dummy variable as the dependent variable. We include the IV along with a set of control variables that are similar to

those used in our main analysis. Table C1 reports estimation results. Our analysis reveals that the IV has a negative and statistically significant effect on a player's decision to participate in social matches each week ($\beta = -.319, p < .05$). In addition, we perform a likelihood ratio test both with and without the instrument to examine potential weak instrument issues. The results of the likelihood ratio test suggest that the model fit improves significantly ($p < .01$) with the inclusion of the IV (Gu and Kannan 2021; Haviv, Huang and Li 2020).

Table C1: Selection into Social Matches—IV Approach

Model	*Full Model*		*Restricted Model*	
Variable	*Estimate*	*S.E.*	*Estimate*	*S.E.*
Instrument $_{it-1}$	-.319**	.128		
Visit frequency$_{it-1}$	1.753***	.020	1.752***	.020
Weekends play portion $_{it-1}$	.058	.037	.059	.037
Night play portion $_{it-1}$	-.174***	.032	-.175***	.032
Age $_{it-1}^2$	-.132***	.041	-.131***	.041
Age $_{it-1}$	.801***	.260	.789***	.260
Tenure $_{it-1}$	.200***	.057	.200***	.057
Precipitation $_{it-1}$	.009	.009	.008	.009
Intercept	-4.588***	.420	-4.565***	.420
User random effects	Included		Included	
Weekly time dummies	Included		Included	
Observations	118,368		118,368	
Log-likelihood	-34,249.434		-34,254.257	
$\Delta 2LL$ (or Deviance)	9.6462***			

Note: *p<.10; **p<.05; ***p<.01. Standard errors are clustered at the user level.

Exclusion. Another essential criterion for a valid instrument is the exclusion restriction, which indicates that the instrument should be correlated with target dependent variable solely through its relationship with the endogenous variable (i.e., social match participation). To test for the exogeneity of our IV, we follow the exclusion test proposed by Barron, Kung, and Proserpio (2021). Specifically, if our instrument is valid, we should observe **no change in purchase activities among players who did not participate in social matches** in response to the instrument. To confirm this, we regressed the weekly purchase amount (i.e., the dependent variable in our main analysis) on the instrument, focusing on the sample of player-week observations without social match participation. From this regression, we find no significant relationship between our instrument and the dependent variable ($\beta = .023; p > .10$). This result demonstrates the validity of our IV as a tool for accounting for the selection process into social matches using solely exogenous variation in the data.

We calculate the inverse Mills ratio (IMR, hereafter) based on the full model presented in Table C1 and include it as an additional regressor in all analyses performed in the manuscript. IMR effectively accounts for the selection bias associated with participating or not participating in

social matches when estimating peer influence. However, we also retain the control function approach, as it accounts for continuous variation in the peer exposure variable, conditional on social match participation.

C.2. First-stage Regression of Peer Exposure (Extent of peer exposure)

To address potential endogeneity of our peer exposure variable, we use the spending of non-overlapping peers—peers of peers whose spending likely influences peers but not necessarily the focal player. Specifically, we regress the endogenous variable, Peer exposure$_{it-1}$, on a set of exogenous variables ($\mathbf{X_{it}}$) and the chosen instrument. Our model is strongly significant and explains a substantial portion of the variance ($F_{9,3698} = 4{,}238.65; p < .001$ and R^2 =72.1%). Consistent with an econometric discussion (Bramoullé, Djebbari and Fortin 2009) and empirical evidence (Shi, Grewal and Sridhar 2021; Sunder, Kim and Yorkston 2019), the average purchases of excluded peers (i.e., peers of peers that focal customers have not interacted with in week *t-2*) significantly increase the purchases of peers in week *t-1* ($\beta = .5003; p < .001$). Following Rossi (2014), we find that the model fit (R^2) increases by 7% upon inclusion of the instrument. We further confirm using the empirical validity test suggested by Sanderson and Windmeijer (2016), that the instrument is not weak (p <.001).

Table C2: Estimation Results of Auxiliary Regression (First-stage Model)

Variable	Estimate	S.E.
Purchases of peers of peers$_{it-2}$ (Instrument)	.500***	.007
Visit frequency$_{it-1}$	.261***	.006
Weekends play portion$_{it-1}$	-.100***	.009
Night play portion$_{it-1}$	-.153***	.007
Age$_{it}^2$	.079	.096
Age$_{it}$	-.427	.504
Tenure$_{it}$	.043**	.013
Precipitation$_{it}$	.001	.002
IMR$_{it-1}$ Selection Correction for Social Match Participation	.268***	.009
Individual Fixed Effects	Included	
Weekly Fixed Effects	Included	
R squared	.694	

Note: N=118,368. Standard errors are clustered at the individual level. Sanderson-Windmeijer (SW) multivariate F test of excluded instrument rejects the null hypothesis that instruments are weak (p<.001) (Sanderson and Windmeijer 2016). In addition, following the suggestion of Rossi (2014), we obtain the R^2 of the first-stage regression models with and without instrument. We confirm that the model fit substantially improves from .674 (without IVs) to .721 (with IVs). *p<.05; **p<.01; ***p<.001.

As with our social match participation model, we examine whether the instrument impacts the focal user's spending solely through its influence on peers' spending, known as the exclusion restriction. Following the approach outlined by Barron et al. (2021), we test the validity of our

instrument using only users who interacted with anonymous peers that had zero spending. If our instrument is valid, we should not observe a significant relationship between the instrument and the focal user's spending for this subset. Consistent with this expectation, we find that the instrument does not significantly influence the focal user's spending (β = .002; p >.10), thereby confirming the exclusion restriction.

WEB APPENDIX D: Full Results of Mechanism Analyses (Table 4)

Table D1: Operationalization of Key Moderators

		Operationalization of	
Table	Model	Contextual Knowledge	Threats to Status
Table D2	(1)	Cum. Game Visits$_{it-1}$	% Inferior match$_{it-1}$ $(R_{it-1} < R_{jt-1})$
Table D2	(2)	Cum. Social Matches$_{it-1}$	% Inferior match$_{it-1}$ $(R_{it-1} < R_{jt-1})$
Table D2	(3)	Tenure$_{it}$	% Inferior match$_{it-1}$ $(R_{it-1} < R_{jt-1})$
Table D3	(4)	Cum. Game Visits$_{it-1}$	% Inferior match$_{it-1}$ $(R_{it-1} + 1 < R_{jt-1})$
Table D3	(5)	Cum. Social Matches$_{it-1}$	% Inferior match$_{it-1}$ $(R_{it-1} + 1 < R_{jt-1})$
Table D3	(6)	Tenure$_{it}$	% Inferior match$_{it-1}$ $(R_{it-1} + 1 < R_{jt-1})$
Table D4	(7)	Cum. Game Visits$_{it-1}$	% Inferior match$_{it-1}$ $(R_{it-1} + 2 < R_{jt-1})$
Table D4	(8)	Cum. Social Matches$_{it-1}$	% Inferior match$_{it-1}$ $(R_{it-1} + 2 < R_{jt-1})$
Table D4	(9)	Tenure$_{it}$	% Inferior match$_{it-1}$ $(R_{it-1} + 2 < R_{jt-1})$

Table D2: Full Results Mechanism Analyses: Model (1)—(3)

Model	(1)		(2)		(3)	
Variables	Est.	S.E.	Est.	S.E.	Est.	S.E.
Peer exposure$_{it-1}$ (PE)	.059***	.012	.074***	.012	.057***	.013
PE × Contextual Knowledge	-.010*	.005	-.006*	.003	-.017**	.006
PE × Threats to Status	-.005	.020	-.002	.020	-.007	.020
Contextual Knowledge	-.067***	.010	-.027***	.006	N/A	
Threats to Status	.182***	.041	.182***	.040	.184***	.043
Visit frequency$_{it-1}$	.170***	.008	.155***	.009	.149***	.009
Weekends play portion$_{it-1}$	.088***	.013	.089***	.014	.092***	.013
Night play portion$_{it-1}$	-.030*	.012	-.033**	.012	-.033**	.012
Age$_{it}^2$	-.212	.185	-.197	.175	-.197	.193
Age$_{it}$	1.140	.958	1.020	.937	1.033	1.015
Tenure$_{it}$	-.071**	.026	-.063*	.027	-.052	.029
Precipitation$_{it}$	.001	.004	.001	.003	.001	.004
IMR$_{it-1}$	-.013	.009	-.014	.010	-.015	.010
Control function$_{it-1}$	-.015	.013	-.025*	.013	-.017	.014
R squared	.279		.279		.278	

Note: N=118,368. All models include individual and weekly time fixed effects. Bootstrapped standard errors with 200 resamples are used. All moderators are mean-centered. *p<.05; **p<.01; ***p<.001.

Table D3: Full Results Mechanism Analyses: Model (4)—(6)

Model	(4)		(5)		(6)	
Variables	Est.	S.E.	Est.	S.E.	Est.	S.E.
Peer exposure$_{it-1}$ (PE)	.066***	.012	.082***	.012	.065***	.012
PE × Contextual Knowledge	-.009*	.004	-.006*	.003	-.015*	.008
PE × Threats to Status	.011	.036	.017	.034	.005	.034
Contextual Knowledge	-.069***	.010	-.027***	.005	N/A	
Threats to Status	.192**	.065	.186**	.065	.192**	.065
Visit frequency$_{it-1}$	.171***	.010	.155***	.010	.150***	.009
Weekends play portion$_{it-1}$	.088***	.015	.089***	.015	.092***	.014
Night play portion$_{it-1}$	-.030*	.012	-.033*	.013	-.033**	.012
Age$_{it}^2$	-.208	.188	-.191	.190	-.193	.185
Age$_{it}$	1.129	1.031	1.003	1.003	1.023	.931
Tenure$_{it}$	-.059*	.028	-.052	.027	-.045	.030
Precipitation$_{it}$	.001	.004	.001	.004	.001	.004
IMR$_{it-1}$	-.009	.010	-.010	.010	-.011	.010
Control function$_{it-1}$	-.023	.013	-.033**	.013	-.026*	.013
R squared	.279		.279		.278	

Note: N=118,368. All models include individual and weekly time fixed effects. Bootstrapped standard errors with 200 resamples are used. All moderators are mean-centered. *p<.05; **p<.01; ***p<.001.

Table D4: Full Results Mechanism Analyses: Model (7)—(9)

Model	(7)		(8)		(9)	
Variables	Est.	S.E.	Est.	S.E.	Est.	S.E.
Peer exposure$_{it-1}$ (PE)	.066***	.011	.082***	.012	.065***	.011
PE × Contextual Knowledge	-.010[a]	.005	-.006*	.003	-.015*	.007
PE × Threats to Status	-.001	.043	.007	.037	-.008	.044
Contextual Knowledge	-.068***	.009	-.027***	.005		
Threats to Status	.233**	.078	.223**	.073	.232**	.077
Visit frequency$_{it-1}$	.172***	.010	.156***	.010	.151***	.009
Weekends play portion$_{it-1}$	.088***	.016	.089***	.015	.093***	.014
Night play portion$_{it-1}$	-.030*	.013	-.033**	.012	-.033**	.012
Age$_{it}^2$	-.207	.172	-.190	.208	-.192	.187
Age$_{it}$	1.128	.922	1.003	1.094	1.024	.973
Tenure$_{it}$	-.059*	.026	-.051	.028	-.044	.026
Precipitation$_{it}$	.001	.004	.001	.003	.001	.003
IMR$_{it-1}$	-.009	.010	-.009	.010	-.011	.009
Control function$_{it-1}$	-.023*	.012	-.033**	.012	-.026*	.012
R squared	.279		.278		.278	

Note: N=118,368. All models include individual and weekly time fixed effects. Bootstrapped standard errors with 200 resamples are used. All moderators are mean-centered. [a]p=.058 *p<.05; **p<.01; ***p<.001.

WEB APPENDIX E:

Robustness Check: Alternative Operationalization of Peer Exposure (Dynamic Perspective)

Players' usage of paid game items may not solely depend on purchases made in the previous week but could also originate from purchases made several weeks prior. To account for this, we construct a dynamically aggregated peer exposure variable, employing a weekly discount factor similar to the advertising stock concept commonly used in marketing literature.

Specifically, we computed the stock of cumulative purchases made by peer j up until week t-1 (see equation E.(1) below).

$$\text{Cum. Purchases } (\$)_{jt-1} = \text{Purchase}(\$)_{jt-1} + \lambda \times \text{Cum. Purchases}(\$)_{jt-2} \qquad \text{E.(1)}$$

where λ is the discount factor for each week, we considered 1.0, 0.7, and 0.7 for robustness. We construct the peer exposure variable by taking the average of the cumulative purchases of each peer that the focal user interacted with. We re-estimated the main models using this newly constructed peer exposure variable in Table E. Across all models in Table E, we find that the results continue to be robust with our theorizing—that anonymous peers exert an influence on focal players and that the primary mechanism is informational influence and not competitive concern, even though the focal context is competitive gaming.

Table E: Estimation Results with Cumulative Purchases Made by Peer *j*

Model	(1)	(2)	(3)
Carryover rate (λ) *for cumulative purchases*	1.0	0.7	0.5
Peer Exposure$_{it-1}$ (PE) *based on cumulative purchases*	.220***	.404***	.438***
	(.048)	(.086)	(.100)
PE$_{it-1}$ × Contextual Knowledge$_{it-1}$	-.011***	-.010***	-.011***
	(.002)	(.003)	(.003)
PE$_{it-1}$ × Threats to Status$_{it-1}$	-.014	-.018	-.018
	(.013)	(.016)	(.018)
Contextual Knowledge$_{it-1}$	-.065***	-.067***	-.067***
	(.010)	(.010)	(.011)
Threats to Status$_{it-1}$	.196***	.201***	.199***
	(.059)	(.052)	(.054)
Visit frequency$_{it-1}$	-.125	-.210*	-.211*
	(.072)	(.088)	(.096)
Weekends play portion$_{it-1}$	.182***	.214***	.215***
	(.028)	(.031)	(.035)
Night play portion$_{it-1}$	.117**	.166***	.169**
	(.040)	(.046)	(.051)
Age$_{it}^2$	-.214	-.202	-.210
	(.170)	(.188)	(.182)
Age$_{it}$	1.212	1.222	1.255
	(.913)	(.966)	(.960)
Tenure$_{it}$	-.109***	-.128***	-.127***
	(.027)	(.029)	(.029)
Precipitation$_{it}$	.002	.001	.001
	(.004)	(.004)	(.004)
IMR$_{it-1}$	-.422***	-.520***	-.515***
	(.099)	(.117)	(.124)
Control function$_{it-1}$	-.187***	-.364***	-.396***
	(.048)	(.086)	(.101)
Individual fixed effects	Included	Included	Included
Weekly fixed effects	Included	Included	Included
R squared	.279	.279	.279

Note: *p<.05; **p<.01; ***p<.001. N=118,368. We use bootstrapped standard errors with 200 resamples. We use cumulative visits as a proxy for contextual knowledge. We also employ the percentage of inferior matches as proxy for threats to status. All moderators are centered on their means.

WEB APPENDIX F:

Robustness Check: Full Results of Cragg's Two-part Hurdle Model

Table F: Cragg-style Two-part Hurdle Model

Dependent variable	$\mathbb{I}(y_{it} > 0)$		$y_{it} \mid y_{it} > 0$	
Model	(1): Purchase Incidence		(2): Purchase Amount	
Variables	Estimate	S.E.	Estimate	S.E.
Peer exposure$_{it-1}$	.022***	.004	-.005	.042
Peer exposure$_{it-1}$ × Know.	-.002[a]	.001	.005	.010
Peer exposure$_{it-1}$× Threat.	-.003	.006	.083	.052
Contextual Knowledge (Know.)	-.020***	.003	-.208***	.044
Threats to Status (Threat.)	.056***	.013	-.087	.109
Visit frequency$_{it-1}$	.058***	.003	.132**	.043
Weekends play portion$_{it-1}$	.031***	.005	.108+	.057
Night play portion$_{it-1}$	-.004	.004	-.043	.055
Age$_{it}^2$	-.039	.065	.500	.794
Age$_{it}$	.237	.362	-3.039	4.670
Tenure$_{it}$	-.023**	.009	.042	.078
Precipitation$_{it}$	.000	.001	-.008	.012
IMR$_{it-1}$	-.007	.004	.025	.034
Control function$_{it-1}$	-.009	.004	-.003	.043
Individual fixed effects	Included		Included	
Weekly fixed effects	Included		Included	
R squared	.225		.498	
Observations	118,368		15,038	

Note: [a]p=.084; *p<.05; **p<.01; ***p<.001. We use bootstrapped standard errors with 200 resamples. We utilized cumulative visit frequency to measure contextual knowledge and the proportion of inferior social matches as indicators of threats to status. All moderators are centered on their means.

WEB APPENDIX G:

Robustness Check: Additional Tests of Competitive Concern

G.1. Analyses based on match outcomes

As we discussed in the manuscript "***Additional test of competitive concern using supplementary data***", the most salient cue for a status threat in the context of video games is likely the outcome of social matches: winning or losing. Simply speaking, if players lose more frequently, it is reasonable to assume that they would experience an increased threat to their status. If this be the case, it would offer ***stronger and more reliable*** support to our hypothesis testing regarding competitive concern. Unfortunately, in the main part of our analyses, we were unable to examine variations based on match results due to data limitations (social match outcomes were not the part of provided data). Specifically, the firm confirmed that they retain match result data for only a limited time frame, and this data was not made available to us.

Second, however, we have found alternative data from another collaborative project with the same data provider that includes information on social match results. This data comes from a different sample and time period but offers a more granular level of user behaviors. One downside to this dataset is that it only captures a four-week period. This is consistent with our previous communication with the firm, which indicates that they do not store match results for more than a month due to data management cost considerations. In Table G1, we describe and compare the two datasets. For the sake of clear communication, we refer to the original dataset as "data used in main analysis" and the new dataset as "supplemental new data".

Table G1: Comparison between Original and New Datasets

Data Scope	*Data used in main analysis*	*Supplemental "new" Data*
Observation period	July 2011 to February 2012	November 2012
Duration	32 weeks or eight months	4 weeks or one month
Login activity	✓	✓
Purchase activity	✓	✓
Social match log	✓	✓
Social match results (win/loss)		✓
Within-game activity logs		
Team management		✓
Mission accomplishment		✓

Since we can observe the outcomes of social matches from the new data, we perform a similar analysis on this dataset, using game results as a potential moderator for competitive concern. This analysis strengthens the manuscript in two ways. First, we examine whether competitive concern is genuinely at work by using the most salient cue for status threat. Second, we replicate our original findings in a different time period while controlling for additional within-game behavior variables (e.g., team management actions and mission accomplishments) that we did

not account for in the main analysis. However, we fully acknowledge and clearly describe in the manuscript that this second dataset is not identical to our main analysis dataset and has a relatively short observation period.

To perform the analyses, we first construct a player-by-week panel dataset and maintain the operationalization of variables exactly as in the main analyses. In addition, we define and add three different operationalizations of 'Threats to Status': **1) Loss rate**, defined as the proportion of lost matches to the total number of social matches played (Mean: .514, Std: .259); 2) **Losses**, defined as the number of lost social matches (Mean: 15.303 Std: 13.729); and 3) **Loss streak**, defined as the maximum consecutive number of lost matches (Mean: 4.856, Std: 5.415). We expect all three variables to serve as good proxies for testing competitive concern as they speak to the extent to which players struggle in the video game. If competitive concern is at work, then we should see the effect of peer exposure increase as threats to status become greater. In terms of empirical setup, we perform a random-effects logistic regression for two reasons. First, we only have two observations per player due to lagging and the use of instruments sourced from the t-2 week. Because of this, including player-level fixed effects would absorb substantial part of the variation in the dataset, rendering hardly any estimate significant. Second, about 2% of observations have a purchase amount greater than zero, making inference on continuous purchase amounts infeasible. We report the results in Table G2.

In Table G2, we continue to find a (1) positive and significant effect of peer exposure and a (2) negative moderating effect of contextual knowledge across all three models presented in the table. However, we do not find a significant moderation effect for the threats to status variable in any of the models. We believe that the analysis using the second dataset ensures the robustness of our findings and provides further assurance that competitive concern is unlikely to be the driver of peer influence in our setting.

Table G2: Mechanism Analyses on Short-term Data: Random Effect Logit Model

Operationalization of Threats to Status Variable	*Loss Rate (%)*		*Lost Matches*		*Loss Streak*	
Model	*(1)*		*(2)*		*(3)*	
Variables	*Est.*	*S.E.*	*Est.*	*S.E.*	*Est.*	*S.E.*
Peer Exposure$_{it-1}$ (PE)	.560**	.176	.473**	.148	.636***	.143
PE × Contextual Knowledge $_{it-1}$	-.111**	.034	-.073*	.035	-.068*	.032
PE × Threats to Status $_{it-1}$	-.320	.221	-.012	.060	-.134	.090
Contextual Knowledge$_{it-1}$	.381***	.049	.406***	.048	.408***	.049
Status Threat$_{it-1}$	-.235	.446	.172	.122	.260	.192
Visit Frequency	.384**	.118	.431***	.126	.424**	.139
Weekends Play Portion$_{it-1}$	.076	.230	.045	.252	.033	.232
Night Play Portion $_{it-1}$	.123	.202	.091	.185	.094	.199
Precipitation $_{it-1}$	-.216	.133	-.208	.140	-.205	.121
In-game managements $_{it-1}$	.110*	.053	.112	.057	.112	.059
In-game mission accomplished $_{it-1}$	.304**	.097	.311***	.091	.317**	.100
IMR $_{it-1}$	-.205*	.103	-.195*	.097	-.217	.115
Control function $_{it-1}$	-.196	.168	-.296	.159	-.338*	.155
Age$_{it}$	.091***	.021	.090***	.021	.090***	.020
Age$_{it}^2$	-.001***	.000	-.001***	.000	-.001***	.000
Male$_{i}$	.198	.128	.195	.130	.198	.141
Social matches played$_{it-1}$	.259**	.093				
Intercept	-8.433***	.669	-8.580***	.609	-8.523***	.663
Variance of random effects (in log)	1.328***	.156	1.323***	.141	1.317***	.155
User-level random effects	Included		Included		Included	
Weekly dummies	Included		Included		Included	
AIC	4368.612		4375.339		4375.667	
BIC	4534.761		4533.180		4533.508	
Observations	29,954		29,954		29,954	

Note: We use bootstrapped standard errors with 200 resamples. Contextual knowledge is measured as cumulative play hours of player i by t-1 week. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

G.2. Suspense as a potential moderator

Insignificant moderation by 'Threats to Status (% inferior matches$_{it-1}$)' might be masked by another factor called suspense (i.e., uncertainty in match outcomes). Users may be more influenced when suspense is high and less influenced when suspense is low. Therefore, even if the competitive effect is present, it can be masked by these two different scenarios.

We explored the alternative approach in this analysis. According to Simonov et al. (2023), suspense can be defined as '… *the standard deviation of the next period's belief about the final outcome of the entertainment product, reflecting viewers' uncertainty about events that may occur*… (page 1-2)'. This implies that when outcome uncertainty is high, suspense increases, and when the outcome becomes pretty certain, suspense decreases.

Using granular observations of the esports video game CS:GO, Simonov et al. (2023) defined the suspense level of streamed content on Twitch.tv for each round t as follows:

$$Suspense_t = \sqrt{\mathrm{E}\left[\sum_w (\tilde{\mu}_{t+1}^w - \tilde{\mu}_t^w)^2 | \mu_t\right]} \; t \in [1, 29] \qquad \text{G.(1)}$$

where μ_t is a belief path and $\tilde{\mu}_t^w$ is observed belief for event $w \in$ {Team A wins, Team B wins}. By construction, the measure increases when '… *uncertainty about the game outcome is high* … (page 8)' (Simonov et al. 2023).

Following this idea, we compute the level of suspense experienced by each player i at week t. The key point of distinction is that Simonov et al. (2023) did not define individual player (or viewer) level suspense, whereas we define it at the individual player level (*i* subscript). Specifically, for each social match played by player i with the opponent player j, we propose that the uncertainty in match outcome (i.e., suspense) will be greater if the *i* and *j* are close to each other in terms of tiers. Contrastingly, the suspense will be lesser if player *i*'s tier is much higher (or lower) than player *j*. To capture these dynamics, we define suspense as follows:

$$\text{Suspense}_{it} = -1 \times \sqrt{E\left[\sum_j \left(Tier_{it} - Tier_{jt}\right)^2\right]} \qquad \text{G.(2)}$$

where $Tier_{it}$ is the observed tier of player i in the game at week t. We multiply minus one to the root mean squared tier difference, as a smaller difference in tiers would suggest greater suspense (i.e., uncertainty in match outcome).

We then introduce this newly measured suspense variable in our main model as a moderator. If the suggested alternative scenario for competitive concern is at work, we should see that the positive and significant peer influence becomes more pronounced when suspense increases. In Table G3, we present the estimation results with this new moderation analysis. Column (1) reports the results when the suspense variable is operationalized as in equation G. (2). Column (2) presents a robustness check when we remove the 'square root' from equation G.(2). As we can see form Table G3, the peer effect is not significantly moderated by suspense. Thus, we conclude that peer influence in our context is more likely driven by uncertainty reduction (i.e.,

informational influence) rather than by competitive concerns (i.e., suspense) experienced during the match.

Table G3: Does Peer Influence Become Stronger as Suspense (Outcome Uncertainty) Increases?

Model	(1)		(2)	
Operationalization of Suspense	Root Mean Squared Tier Difference		Mean Squared Tier Difference	
Variables	Est.	S.E.	Est.	S.E.
Peer exposure$_{it-1}$ (PE)	.059***	.013	.069***	.012
PE × Suspense$_{it-1}$	.0020	.0023	-.0001	.0003
Suspense$_{it-1}$	-.023***	.004	-.002**	.001
Visit frequency$_{it-1}$	.144***	.010	.148***	.010
Weekends play portion$_{it-1}$	.095***	.015	.095***	.015
Night play portion$_{it-1}$	-.028*	.014	-.030*	.014
Age$_{it}^2$	-.204	.178	-.207	.178
Age$_{it}$	1.063	.924	1.077	.936
Tenure$_{it}$	-.109***	.025	-.110***	.026
Precipitation$_{it}$	.001	.004	.001	.004
IMR$_{it-1}$	-.023*	.011	-.015	.010
Control function$_{it-1}$	-.020	.013	-.031*	.013
Individual fixed effects	Included		Included	
Weekly fixed effects	Included		Included	
R squared	.278		.278	

Note: N=118,368. We use bootstrapped standard errors with 200 resamples. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

Finally, for robustness, we also consider relatively simpler operationalizations of suspense variables by just computing the number of social matches with opponents (1) with exactly the same tier and (2) with relatively similar tiers (± two tier differences) as alternative moderators. The expectation being that matches with tighter tier differences would create more suspense due to unpredictable match outcomes. We did not find significant moderation in these variables either, and the results are listed in Table G4.

Table G4: Does Peer Influence Become Stronger as Suspense (Outcome Uncertainty) Increases?

Model	(1)		(2)	
Operationalization of Suspense	Matches with exactly same tier users		Matches with similar tier (± 2 tiers) users	
Variables	Est.	S.E.	Est.	S.E.
Peer exposure$_{it-1}$ (PE)	.049***	.013	.033*	.013
PE × Suspense$_{it-1}$	-.0007	.0005	-.0004	.0003
Suspense$_{it-1}$	.004**	.001	.003***	.001
Visit frequency$_{it-1}$	.151***	.010	.151***	.009
Weekends play portion$_{it-1}$	.097***	.015	.099***	.016
Night play portion$_{it-1}$	-.031*	.014	-.031**	.011
Age$_{it}^2$	-.215	.182	-.217	.186
Age$_{it}$	1.120	.959	1.123	.954
Tenure$_{it}$	-.109***	.024	-.109***	.025
Precipitation$_{it}$	.001	.003	.001	.003
IMR$_{it-1}$	-.003	.010	.002	.010
Control function$_{it-1}$	-.013	.013	-.002	.014
Individual fixed effects	Included		Included	
Weekly fixed effects	Included		Included	
R squared	.278		.278	

Note: N=118,368. We use bootstrapped standard errors with 200 resamples. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

G.3. Mid-status anxiety

Another possibility is that peer influence with competitive concern may be non-linear, as suggested by Hu and Van den Bulte (2014). Peer influence may be strongest for those in middle status due to mid-status anxiety. We used the weekly recorded tier of each player to indicate their status and included both quadratic and linear terms of the status variable, interacting them with the peer exposure variable. If middle-status anxiety, a component of competitive concern, is present, the impact of peer exposure should be moderated in an inverted U-shaped manner by one's status. As reported in Table G5, we do not find that one's status or its square term significantly moderates the positive effect of peer exposure.

Table G5: Analysis for Mid-Status Anxiety following Hu and Van den Bulte (2014)

Variables	Estimate	S.E.
Peer exposure$_{it-1}$ (PE)	.068***	.012
Own status$_{it-1}$	.00084	.00128
Own status$_{it-1}^2$	-.00036	.00022
PE × Own status$_{it-1}$	.00039	.00106
PE × Own status$_{it-1}^2$	.00019	.00012
Visit frequency$_{it-1}$	.150***	.009
Weekends play portion$_{it-1}$	.094***	.013
Night play portion$_{it-1}$	-.032**	.011
Age$_{it}^2$	-.209	.200
Age$_{it}$	1.088	1.069
Tenure$_{it}$	-.094**	.030
Precipitation$_{it}$	.001	.003
IMR$_{it-1}$	-.010	.009
Control function$_{it-1}$	-.032**	.012
Individual fixed effects	Included	
Weekly fixed effects	Included	
R squared	.278	

Note: N=118,368. We use bootstrapped standard errors with 200 resamples. The moderator is centered on its mean. *p<.05; **p<.01; ***p<.001.

WEB APPENDIX H:

Robustness Check: Alternative Sampling Criteria

We confirm the robustness of our findings using alternative samples. First, we replicate the analyses with the original 4,000 players. Since approximately 7.5% of users have missing control variables, we exclude these variables from our replication. As shown in Table H1 below, the positive effect of peer exposure is replicated; however, it attenuates as one's contextual knowledge increases, supporting the theory of informational influence. We use cumulative visits as a proxy for contextual knowledge. We also employ the percentage of inferior matches (where the player's tier is lower than that of the opponents) in a manner consistent with our main analyses for mechanism tests.

Table H1: Replication with Original 4,000 Players Sample

Variables	Est.	S.E.
Peer Exposure$_{it-1}$	.056***	.012
PE × Contextual Knowledge	-.013***	.004
PE × Threats to Status	.004	.017
Contextual Knowledge$_{it-1}$	-.068***	.010
Threats to Status	.156***	.036
Visit frequency$_{it-1}$	.180***	.009
Weekends play portion$_{it-1}$	.085***	.013
Night play portion$_{it-1}$	-.038**	.013
Tenure$_{it}$	-.057*	.025
IMR$_{it-1}$	-.007	.010
Control function$_{it-1}$	-.012	.013
Individual Fixed Effects	Included	
Weekly Fixed Effects	Included	
R squared	.283	

Note: Observations is 128,000. We use bootstrapped standard errors with 200 resamples. We utilized cumulative visit frequency to measure contextual knowledge and the proportion of inferior social matches as indicators of threats to status. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

Next, we replicate the analyses with the top 1,131 players, who account for 95% of the firm's revenues. This conditioning is done to exclude players who never contribute to the firm's revenues under any circumstances. Such 'always-free' users are common in other freemium-based services. As shown in Table H2, we continue to find evidence of peer exposure and the negative moderation of contextual knowledge. Contextual knowledge and threats to status variables are the same as Table H1.

Table H2: Replication with Top 1,131 Players Who Account for 95% of Revenues

Variables	Est.	S.E.
Peer Exposure$_{it-1}$	.081*	.034
PE × Contextual Knowledge	-.031***	.009
PE × Threats to Status	-.049	.051
Contextual Knowledge	-.151***	.029
Threats to Status	.410***	.109
Visit frequency$_{it-1}$	.371***	.029
Weekends play portion$_{it-1}$	.201***	.039
Night play portion$_{it-1}$	-.075*	.035
Age$_{it}^2$	-.808	1.078
Age$_{it}$	3.947	6.752
Tenure$_{it}$	-.197**	.069
Precipitation$_{it}$	.012	.011
IMR$_{it-1}$	.004	.031
Control function$_{it-1}$	-.015	.034
Individual Fixed Effects	Included	
Weekly Fixed Effects	Included	
R squared	.249	

Note: Observations is 36,192. We use bootstrapped standard errors with 200 resamples. We utilized cumulative visit frequency to measure contextual knowledge and the proportion of inferior social matches as indicators of threats to status. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

WEB APPENDIX I:

Robustness Check: Test for Normative Influence

To empirically examine normative influence (Deutsch and Gerard 1955) as a possible mechanism for the identified peer effect, we perform additional analyses. Since the need for conformity to social norms is known to increase as one's peer network expands (Iyengar, Van den Bulte and Lee 2015; Toker-Yildiz et al. 2017; Zhang, Liu and Chen 2015), we investigate whether the magnitude of the estimated peer effect varies with the size of the social network. We use the number of unique social matches and the number of unique peers with whom players have interacted as proxies for social network size. Cumulative visits serve as a proxy for contextual knowledge. Additionally, we employ the percentage of inferior matches (where the player's tier is lower than that of the opponents) as a proxy for threats to status in a manner consistent with our main analyses for mechanism tests.

In Table I, we continue to observe evidence of informational influence through the negative and significant moderation of contextual knowledge; however, we find no evidence supporting normative influence.

Table I: Test for Normative Influence as a Possible Underlying Mechanism

Operationalization of **Social Network Size**		(1) Number of Social Matches$_{it-1}$		(2) Number of Stranger Peers Interacted$_{it-1}$	
Variables	Test for	Est.	S.E.	Est.	S.E.
Peer Exposure$_{it-1}$		.003	.013	.002	.013
PE × Contextual Knowledge	Informational Influence	-.013**	.004	-.013**	.005
PE × Threats to Status	Competitive Concern	-.007	.021	-.007	.021
PE × Social Network Size	Normative Influence	-.0001	.000	-.0002	.000
Contextual Knowledge		-.069***	.010	-.069***	.011
Threats to Status		.183***	.041	.183***	.041
Social Network Size		.004***	.001	.004***	.001
Visit frequency$_{it-1}$		.167***	.010	.167***	.009
Weekends play portion$_{it-1}$		.094***	.014	.093***	.015
Night play portion$_{it-1}$		-.028*	.013	-.028*	.011
Age$_{it}^2$		-.211	.175	-.208	.184
Age$_{it}$		1.125	.891	1.115	.941
Tenure$_{it}$		-.074**	.027	-.074**	.027
Precipitation$_{it}$		.001	.003	.001	.004
IMR$_{it-1}$		.005	.010	.005	.010
Control function$_{it-1}$		.028*	.014	.029*	.013
Individual Fixed Effects		Included		Included	
Weekly Fixed Effects		Included		Included	
R squared		.280		.280	

Note: Observations: 118,368. We use bootstrapped standard errors with 200 resamples. We utilized cumulative visit frequency to measure contextual knowledge and the proportion of inferior social matches as indicators of threats to status. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

WEB APPENDIX J:

Robustness Check: Alternative Functional Form (Poisson Fixed Effects Model)

To account for the frequent occurrence of non-purchases in free-to-play games, we re-estimate the main model with a Poisson fixed effect model using purchase frequency as a dependent variable (Wang and Goldfarb 2017). We employ the percentage of inferior matches—where the player's tier is lower than that of the opponents—as a proxy for threats to status. As proxies for contextual knowledge, we use cumulative game visits. In Table J, we find that the impact of peers diminishes as one's contextual knowledge increases. However, the main effect of peer exposure is insignificant. Since the high-order interaction term yields a significant and consistent effect, we consider the results to still be within the range of robustness.

Table J: Poisson Fixed Effects Model

Variables	Est.	S.E.
Peer Exposure$_{it-1}$	.100	.080
PE X Contextual Knowledge	-.052**	.018
PE X Threats to Status	-.085	.089
Contextual Knowledge	-.370***	.083
Threats to Status	.494*	.233
Visit frequency$_{it-1}$	.813***	.073
Weekends play portion$_{it-1}$	.520***	.136
Night play portion$_{it-1}$	.164	.127
Age$_{it}^2$	-.732	1.430
Age$_{it}$	3.717	8.259
Tenure$_{it}$	-.216	.166
Precipitation$_{it}$	.020	.026
IMR$_{it-1}$	.070	.079
Control function$_{it-1}$	-.064	.084
Individual Fixed Effects	Included	
Weekly Fixed Effects	Included	
Pseudo R squared	.614	

Note: Observations is 118,368. We use bootstrapped standard errors with 200 resamples. Cumulative visit frequency is used to measure contextual knowledge, and the proportion of inferior social matches serves as an indicator of threats to status. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

WEB APPENDIX K:

Robustness Check: Need for Reloading Game Items

Table K: Control for Reloading of Game Items

Variables	Est.	S.E.
Peer Exposure$_{it-1}$	.170***	.010
PE × Contextual Knowledge	-.013**	.004
PE × Threats to Status	-.028	.021
Contextual Knowledge	-.022*	.009
Threats to Status	.241***	.044
Social matches participated$_{it-1}$	.003***	.000
Weekends play portion$_{it-1}$	.164***	.014
Night play portion$_{it-1}$	.073***	.010
Age$_{it}^2$	-.206	.173
Age$_{it}$	1.112	.931
Tenure$_{it}$	-.085**	.027
Precipitation$_{it}$	.001	.004
IMR$_{it-1}$	-.131***	.008
Control function$_{it-1}$	-.138***	.011
Individual Fixed Effects	Included	
Weekly Fixed Effects	Included	
R squared	.277	

Note: Observations is 36,192. We use bootstrapped standard errors with 200 resamples. We utilized cumulative visit frequency to measure contextual knowledge and the proportion of inferior social matches as indicators of threats to status. All moderators are centered on their means. *p<.05; **p<.01; ***p<.001.

WEB APPENDIX L: Heterogeneity in Item Salience

In addition to the analyses reported on product salience heterogeneity, we explore the heterogeneity in what the focal users purchase using the same information. Specifically, we compute the ratio of the purchase amount ($) for highly salient items to the total purchase amount ($) +1 as follows:

$$\frac{\left(Purchase(\$)_{it}^{High\ Salience}\right)}{\left(Purchase(\$)_{it}^{High\ Salience} + Purchase(\$)_{it}^{Low\ Salience} + 1\right)} \qquad \text{L.(1)}$$

The proposed ratio-style dependent variable would capture the relative share of focal player's item purchases with greater salience to the total purchases made in week t. To prevent division by zero, one is added to the denominator. We replicate the same analysis we performed for item salience with this new dependent variable. Table L reports the estimation results.

Table L: Peer Influence by Item Salience

Variables	Est.	S.E.
Peer exposure$_{it-1}^{High\ Salience}$ $(PE: High)$	.032***	.005
Peer exposure$_{it-1}^{Low\ Salience}$ $(PE: Low)$	-.057***	.014
Visit frequency$_{it-1}$	.038***	.002
Weekends play portion$_{it-1}$	.016***	.003
Night play portion$_{it-1}$	-.010***	.003
Age$_{it}^2$	-.035	.049
Age$_{it}$	.185	.263
Tenure$_{it}$	-.017*	.007
Precipitation$_{it}$	.001	.001
IMR$_{it-1}$	.004	.003
Control Function$_{it-1}$ (PE:High)	-.026***	.005
Control Function$_{it-1}$ (PE:Low)	.064***	.014
Individual fixed effects	Included	
Weekly fixed effects	Included	
R squared	.288	

Note: Observations is 118,368. We use bootstrapped standard errors with 200 resamples. *p<.05; **p<.01; ***p<.001.

We found a positive and significant influence of peer exposure to high salience items (β= .032, p<.001), meaning more exposure to peers' use of salient game items increases the share of the same kinds of items among purchased items. We also found a negative and significant influence of peer exposure to low salience items (β= -.057, p<.001). This is consistent with our theorization that when focal users are exposed to peers' uses of low-saliency items, the peer influence becomes weaker, leading to a drop in the ratio of salient item purchases. This does not necessarily mean that focal players buy more low-salience items; rather, it indicates a diminished

influence from peers on the purchase of salient items. The negative effect of low-salience items could also be partly driven by instances where the focal player decides not to purchase anything (i.e., reduced peer influence), resulting in the dependent variable becoming zero.

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